Sensitivity Analysis of Key Assumptions – With Profit Endowment Assurance

Introduction

Sensitivity analysis was conducted to evaluate the impact of changes in key pricing and valuation assumptions on the profitability of the with-profit endowment assurance product. The analysis focuses on three critical variables: premium, sum assured, and interest rate. By varying one assumption at a time while keeping all other assumptions constant, the analysis identifies the extent to which each variable influences profitability and highlights the potential risks associated with adverse changes in these assumptions.

1. Sensitivity Analysis of Premium

Observations

- Profitability increases as the premium level increases.
- At a premium of ₹22,000, the policy generates a loss of ₹46,321.87.
- Losses gradually reduce as premiums increase from ₹22,000 to ₹25,000.
- The policy reaches the break-even point between ₹25,000 and ₹26,000, where profit changes from -₹10,709.23 to ₹1,161.65.
- Profit continues to increase steadily beyond the break-even level, reaching ₹60,516.04 at a premium of ₹31,000.
- The relationship between premium and profit appears positive and approximately linear across the range tested.

Implications

The results indicate that premium is a key determinant of profitability for the with-profit endowment assurance product. Higher premiums generate greater income for the insurer, thereby improving profit margins and strengthening the financial viability of the product. Conversely, premiums set below the break-even level are insufficient to cover expected benefits, expenses, and reserve requirements, resulting in losses. This highlights the importance of accurate premium pricing and regular reviews of premium adequacy to ensure the product remains profitable throughout its duration.

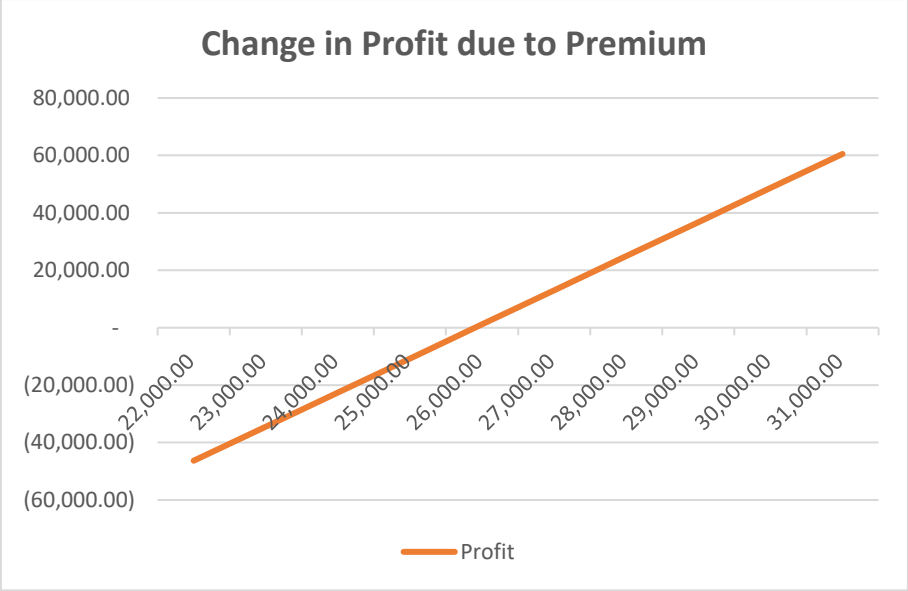


Figure 1: Sensitivity of Profit to Changes in Premium Levels

2. Sensitivity Analysis of Sum Assured

Observations

- Profitability decreases as the sum assured increases.
- At a sum assured of ₹300,000, the profit is ₹123,230.44.
- Profit declines consistently as the sum assured increases from ₹300,000 to ₹750,000.
- The profit reduces to approximately ₹1,161.65 at a sum assured of ₹500,000.
- Beyond ₹500,000, the policy becomes unprofitable, with losses increasing as the sum assured rises.
- At a sum assured of ₹750,000, the model records a loss of ₹151,424.34.
- The relationship between sum assured and profit is strongly negative.

Implications

The analysis demonstrates that the sum assured has a significant impact on profitability because it directly determines the magnitude of future benefit payments. As the level of coverage increases, the insurer's expected liability also increases, reducing profit when premiums remain unchanged. This finding highlights the need for insurers to ensure that premium rates adequately reflect the benefits being provided. Failure to appropriately align premiums with higher sums assured may result in substantial losses and threaten the long-term sustainability of the product.

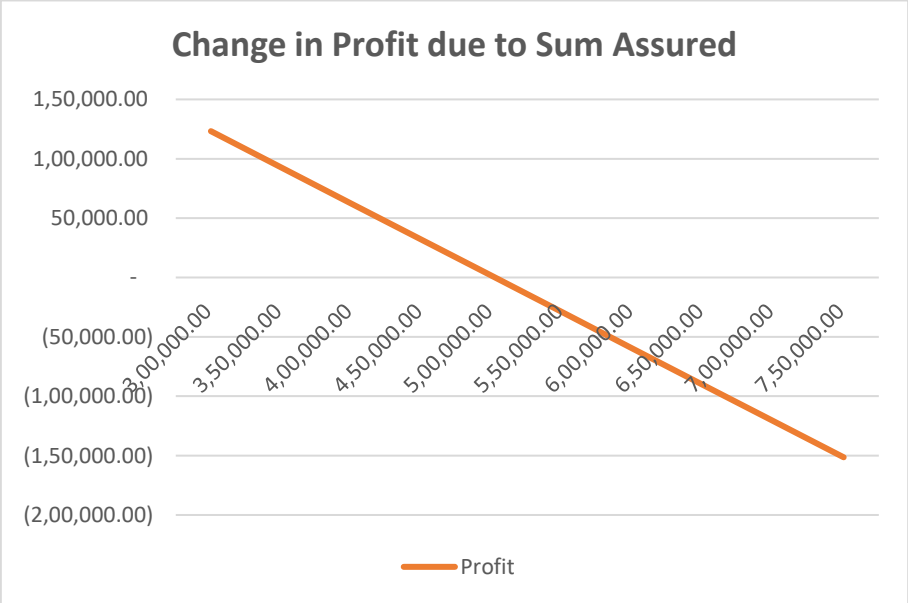


Figure 2: Sensitivity of Profit to Changes in Sum Assured

3. Sensitivity Analysis of Interest Rate

Observations

- Profitability increases as the interest rate assumption increases.
- At an interest rate of 2.00%, the policy records a loss of ₹145,384.20.
- Losses reduce steadily as the interest rate increases.
- The model reaches the break-even point around the base assumption of 5.00%, where profit is approximately ₹1,161.65.
- Profit continues to rise beyond this level, reaching ₹80,247.96 at an interest rate of 8.75%.
- The relationship between interest rate and profitability is positive throughout the range analysed.

Implications

Interest rate assumptions significantly influence the valuation of long-term insurance liabilities. Higher interest rates reduce the present value of future benefit obligations, thereby improving profitability. Conversely, lower interest rates increase the present value of liabilities and can result in substantial losses. The findings highlight the importance of prudent investment management and realistic interest rate assumptions when pricing and valuing life insurance products. Since investment returns are subject to economic and market conditions, insurers must regularly monitor interest rate risk and maintain appropriate asset-liability matching strategies.

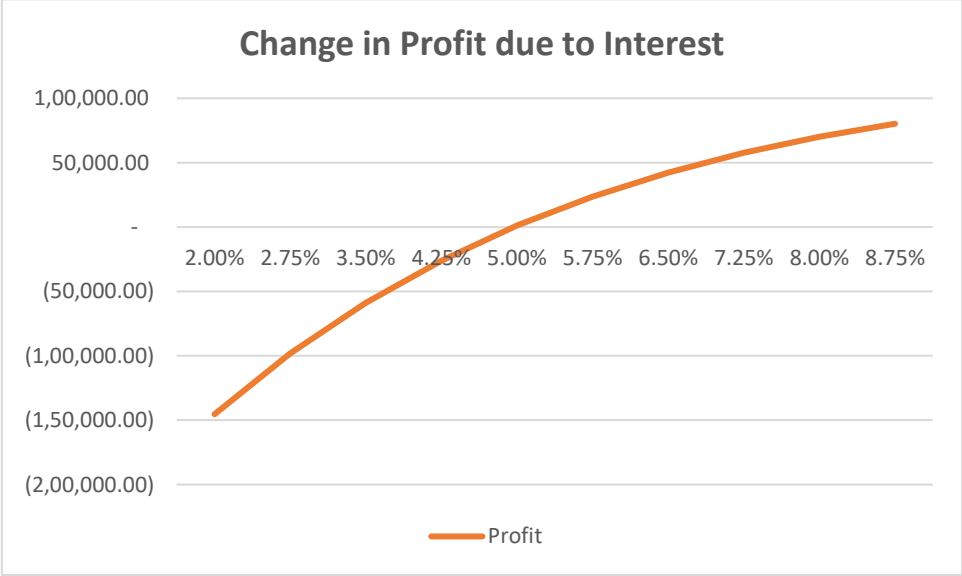


Figure 3: Sensitivity of Profit to Changes in Interest Rate Assumptions

Overall Findings

The sensitivity analysis confirms that profitability is highly dependent on the assumptions used in the valuation model. Premium and interest rate assumptions exhibit a positive relationship with profitability, whereas the sum assured exhibits a negative relationship. Among the variables analysed, changes in the sum assured and interest rate produce the largest variations in profit, indicating that the model is particularly sensitive to assumptions affecting policy liabilities and discounting. The analysis also identifies clear break-even points for each assumption, beyond which profitability changes significantly. Overall, the results emphasise the importance of selecting appropriate pricing and valuation assumptions, as even moderate changes can materially affect the financial performance and sustainability of the with-profit endowment assurance product.

Assumption

The base assumptions used in this valuation model were derived from the *Institute of Actuaries of India (IAI) CM1B – Actuarial Mathematics Practice Examination Paper, September 2021*. Selected assumption values were varied for sensitivity analysis purposes while maintaining the original modelling framework. In addition, death benefits were assumed to be payable at the end of the year of death.